

BIOMECHANICS (EPHE 341)

# Angular Kinetics 2

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Centre of gravity: where it is, how anyone found out, and why every turning problem in the course has been quietly about it.

# Angular Kinetics 2

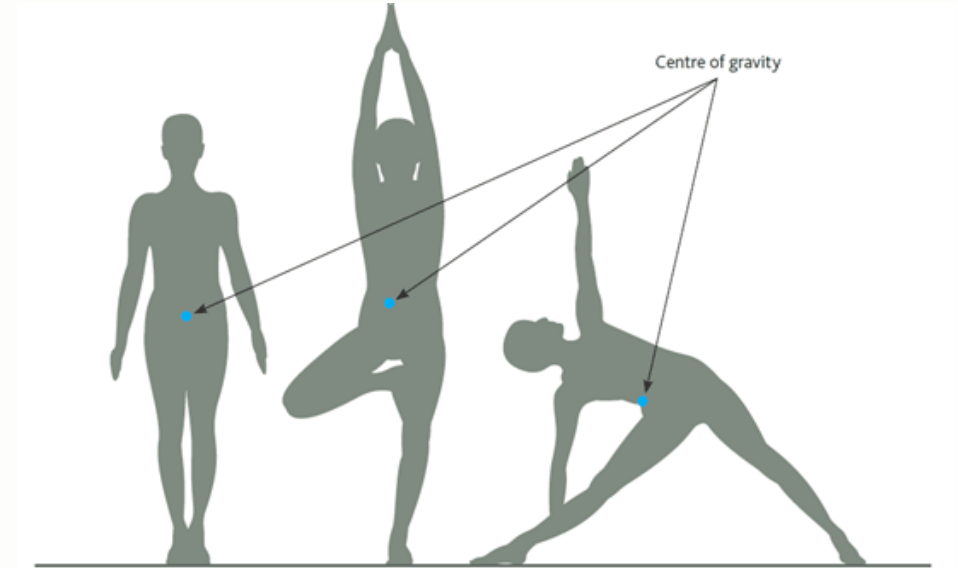
## *Centre of mass, centre of gravity*

**Centre of mass** — the average location of all of the mass in a body.

**Centre of gravity** — the point about which the weight is evenly distributed, and through which the force of gravity acts.

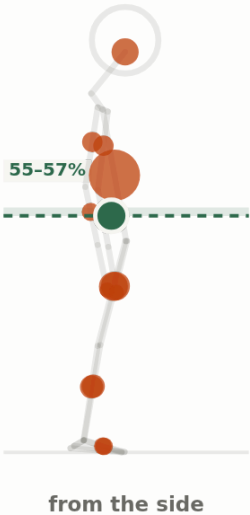
For anything near the earth's surface the two coincide, because the gravitational field is the same strength everywhere in the body. The deck uses **CofG** throughout, as his slides do.

They are not the same idea. Centre of mass is about where the matter is; centre of gravity is about where the *weight* acts. Take an object tall enough that  $g$  is measurably weaker at the top and the two separate — which matters for a rocket and never for a person.



The point moves when the body does — and it does not have to stay inside the body at all.

Is it really 55 to 57% of standing height?



adding the segments, feet upwards

|             |        |          |         |
|-------------|--------|----------|---------|
| Foot R      | 0.0145 | 0.80 kg  | 2.3 cm  |
| Foot L      | 0.0145 | 0.80 kg  | 2.3 cm  |
| Leg R       | 0.0465 | 2.56 kg  | 16.1 cm |
| Leg L       | 0.0465 | 2.56 kg  | 19.3 cm |
| Thigh R     | 0.1000 | 5.50 kg  | 38.7 cm |
| Thigh L     | 0.1000 | 5.50 kg  | 46.0 cm |
| Hand R      | 0.0060 | 0.33 kg  | 46.3 cm |
| Hand L      | 0.0060 | 0.33 kg  | 46.5 cm |
| Forearm R   | 0.0160 | 0.88 kg  | 48.5 cm |
| Forearm L   | 0.0160 | 0.88 kg  | 50.3 cm |
| Upper arm R | 0.0280 | 1.54 kg  | 55.0 cm |
| Upper arm L | 0.0280 | 1.54 kg  | 59.0 cm |
| Trunk       | 0.4970 | 27.34 kg | 83.2 cm |
| Head + neck | 0.0810 | 4.46 kg  | 88.6 cm |

so far 1.0000 55.00 kg 88.6 cm

whole body: 88.6 cm of 161 cm = 55.1% of stature

this slide says 55 to 57% — and the fractions sum to 1.000, so nothing has been left out

Measured · Subject16 · 161 cm, 55 kg · Winter’s table on this person’s own standing marker set · the head-and-neck end point is estimated, which moves the answer by half a point

Segments  
added



From the side

From behind

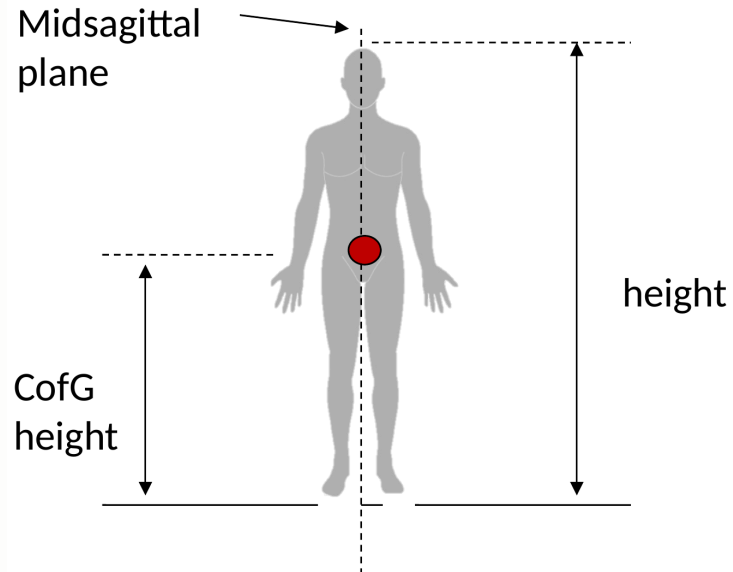
Subject16

Subject15

▶ Add them up

Drag the slider and watch the answer climb from the feet. The **trunk** is the moment that matters — half the body mass arriving at once. Both subjects finish inside his 55 to 57% band, and the mass fractions add to 1.000, so this is the whole body and not a part of it.

## *Where it sits, and why the midline*



In the anatomical position the centre of gravity is **55–57%** of standing height — a little above the navel, roughly at the level of the second sacral vertebra.

Because the body is close to symmetrical left and right, it lies in the **midsagittal plane**, the plane that divides the body into left and right halves.

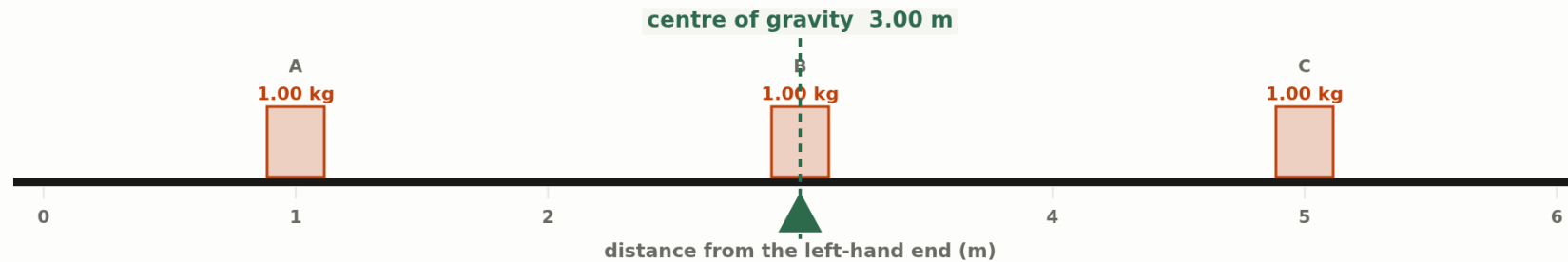
Symmetry is doing the work in that second sentence, not anatomy. Any body whose mass is mirrored about a plane has its centre of gravity in that plane, whatever shape it is. Break the symmetry — lift one arm, carry a bag — and the point leaves the midline immediately.

# Finding it

One sum, and the reason that sum is the right one

# Angular Kinetics 2

## The weighted average, and the balance point



$$\begin{aligned}\text{CofG} &= \Sigma(m \cdot r) \div \Sigma m \\ &= ( (1)(1) + (1)(3) + (1)(5) ) \div ( 1 + 1 + 1 ) \\ &= 3.00 \text{ m}\end{aligned}$$

drag a block along the bar · the answer is the mean position weighted by mass

Three equal masses

Stacked

Thigh, leg, foot

The formula

The torques about it

The centre of gravity is the **mean position of the mass**. Spread the three equal masses out or stack them in one place and the answer is the same 3 m, which is the clue that only the *weighted mean* matters. Then press **The torques about it**: the turning effects of the three weights cancel at that point and nowhere else. That is why it is the balance point.

## *Why that sum is the balance point*

Gravity pulls on every particle of a body. Each particle's weight makes a moment about any point you care to choose.

The centre of gravity is the one point for which all of those moments **cancel**:

$$\sum m_i g (r_i - r_{\text{cofg}}) = 0$$

Rearranging that, and cancelling the  $g$  that appears in every term:

$$r_{\text{cofg}} = \sum (m_i r_i) \div \sum m_i$$

The  $g$  cancels, which is why the centre of gravity is in the same place on the Moon — and why “centre of mass” and “centre of gravity” give the same answer here.

So the whole weight of the body can be replaced by a single force  **$W = mg$**  acting at that one point, and every moment it makes about every other point comes out right.

That substitution is what makes the rest of the course possible. A free body diagram of a limb draws one weight vector, not a cloud of them.

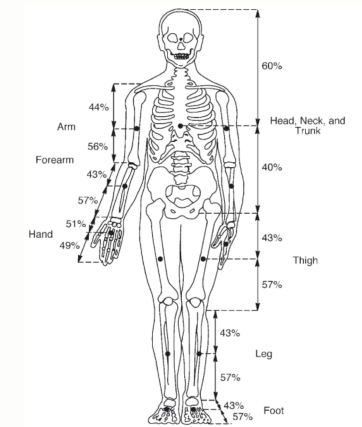
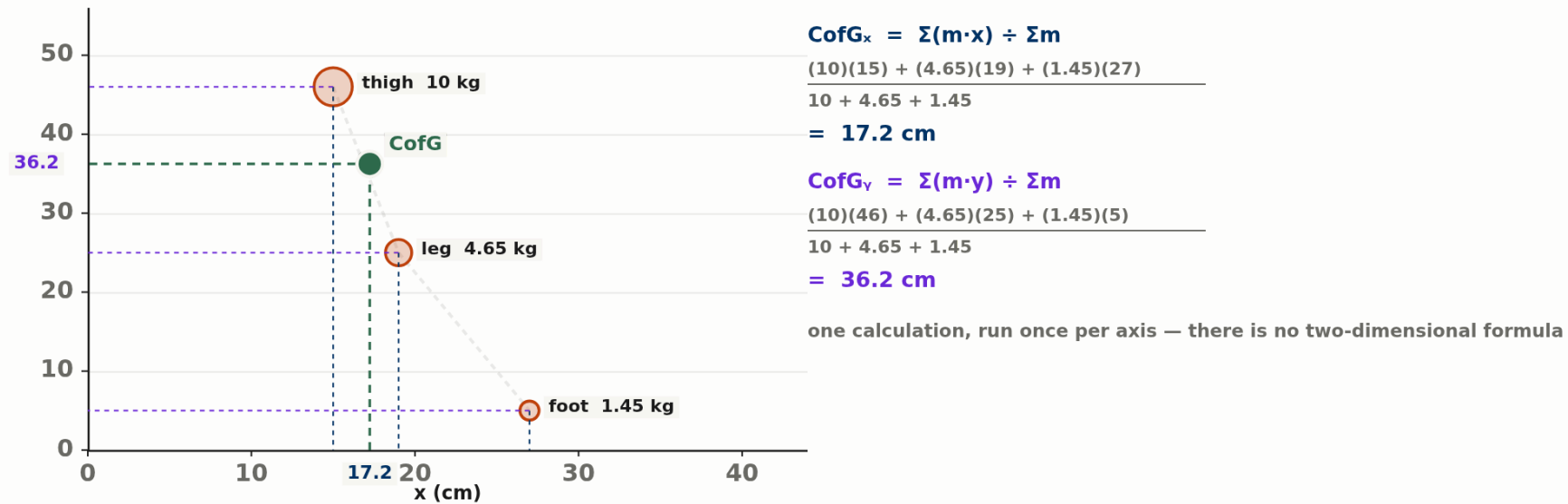


FIGURE 1.15. Estimated Body Segment Centers of Gravity Expressed as a Percentage of Segment Length (after Dempster 1955; Williams and Lissner 1962)

# Angular Kinetics 2

*Two dimensions is the same sum, twice*



Measured · the masses are Winter's fractions for a 100 kg person: thigh 0.100, leg 0.0465, foot 0.0145

Both axes

x only

y only

Reset to his numbers

Drag a segment. The x sum and the y sum are the **same calculation** run on different coordinates, and neither one knows about the other — which is why the centre of gravity of a limb can sit outside the limb entirely. Bend it far enough and watch the green dot leave the leg.

# Where the numbers come from

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Segment masses and segment centres of gravity, and the four ways anyone has ever measured them

## Segment centre of gravity

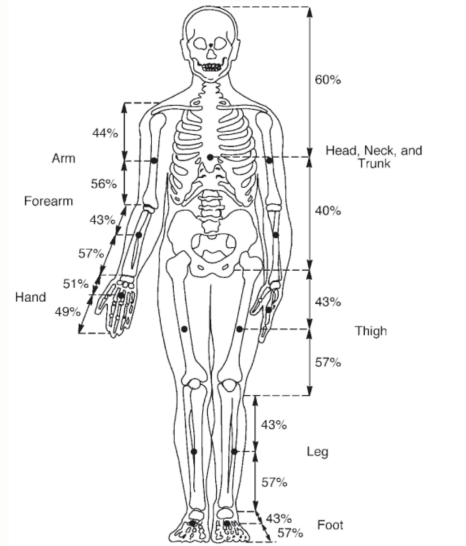


FIGURE 1.15. Estimated Body Segment Centers of Gravity Expressed as a Percentage of Segment Length (after Dempster 1955; Williams and Lissner 1962)

To find the centre of gravity of a *whole body* you need the mass and the centre of gravity of every **segment**.

For a living person neither can be measured directly. Four routes have been taken:

1. **Reaction board** — a plank, a knife edge and a scale;
2. **Cadaver dissection** — segment, weigh, balance;
3. **Photogrammetry** — photographs, volumes, assumed densities;
4. **Scanning** — DEXA and MRI, which measure tissue directly.

The percentages on the figure are remarkably consistent: the centre of gravity of most limb segments sits at about **43%** of the way from the proximal end. Limbs taper, so the mass is nearer the joint it hangs from.

**UVIC** EPHE 341 · BIOMECHANICS

The diagram illustrates a horizontal line representing a thigh segment. The left end is labeled 'proximal' and the right end is labeled 'distal'. A green dot on the line represents the 'centre of gravity'. Above the line, a blue bracket indicates the distance from the proximal end to the center of gravity, labeled 'Thigh · Greater trochanter / femoral condyles' and '0.433 L = 18.0 cm'. Below the line, a purple bracket indicates the total distance from the proximal end to the distal end, labeled 'segment length 41.6 cm' and '0.567 L = 23.6 cm'.

**10.0 kg**

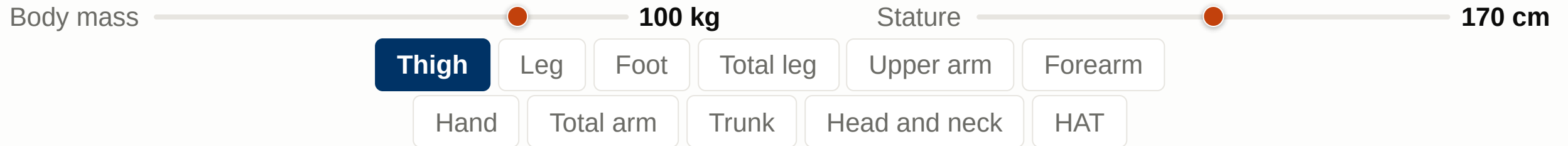
**0.181 kg·m<sup>2</sup>**

**0.506 kg·m<sup>2</sup>**

**0.740 kg·m<sup>2</sup>**

at 100 kg this is the thigh mass on his slide 8

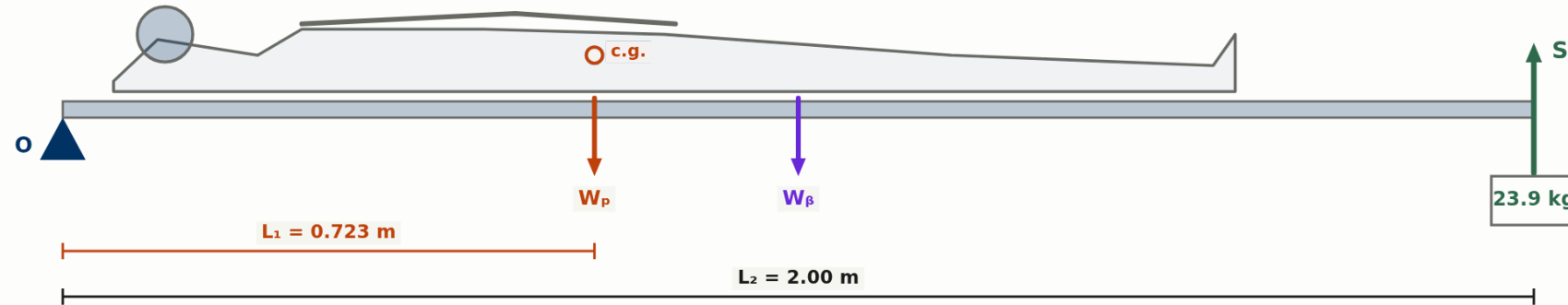
Winter, Table 4.1, after Dempster — the table on his slide 15, with a body put into it



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# Angular Kinetics 2

## The reaction board



$$\Sigma M \text{ about } O = 0 \rightarrow S \cdot L_2 - W_p \cdot L_1 - W_b \cdot \frac{1}{2} L_2 = 0$$

$$S = (540 \times 0.723 + 78 \times 1.00) \div 2.00 = 234 \text{ N} = 23.9 \text{ kg on the scale}$$

and read backwards:  $L_1 = (S \cdot L_2 - W_b \cdot \frac{1}{2} L_2) \div W_p = 0.723 \text{ m from the head} \rightarrow \text{centre of gravity at 55.1\% of stature}$

Measured · Subject16 · 161 cm, 55 kg · his slide writes the three terms all added, which cannot be zero — the scale pushes up while the weights pull down

Board length

2.00 m

Board mass

8.0 kg

Arms at the side

Arms overhead

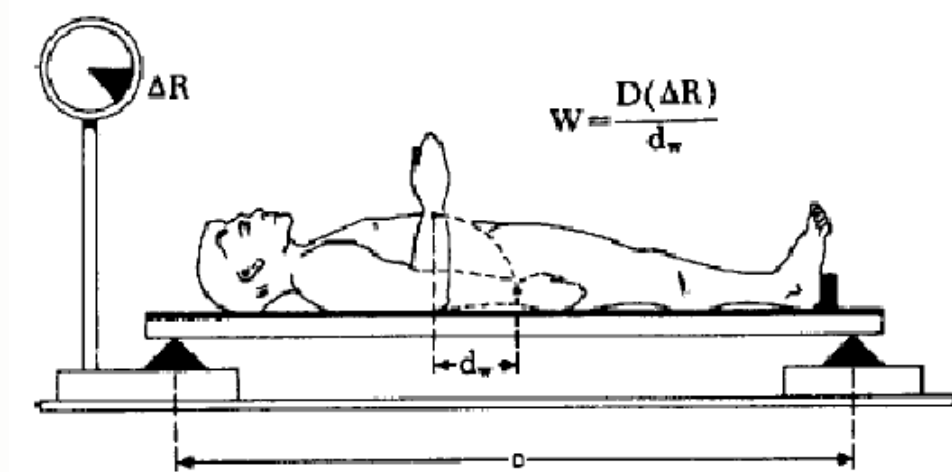
Subject16

Subject15

One scale reading locates the centre of gravity of a whole person. Raise the arms overhead and the centre of gravity moves **toward the head**, which is the pivot, so the lever arm shortens and the scale reads **less** — the same weight, measured the same way, and a different answer because the body changed shape.

# Angular Kinetics 2

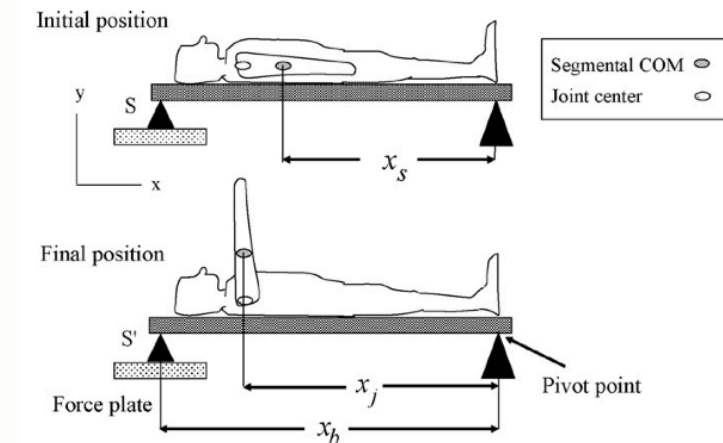
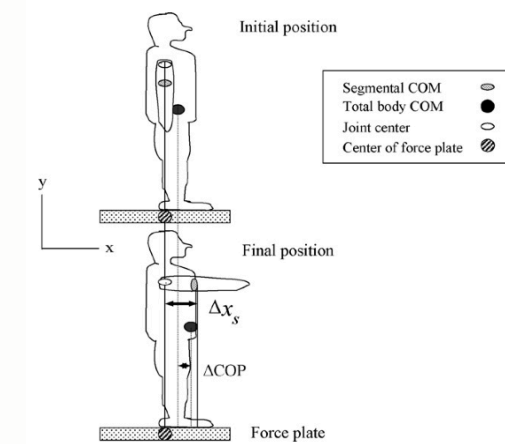
*The same board, one segment at a time*



Move one segment a known distance, read the change in the scale, and the mass of that segment follows:

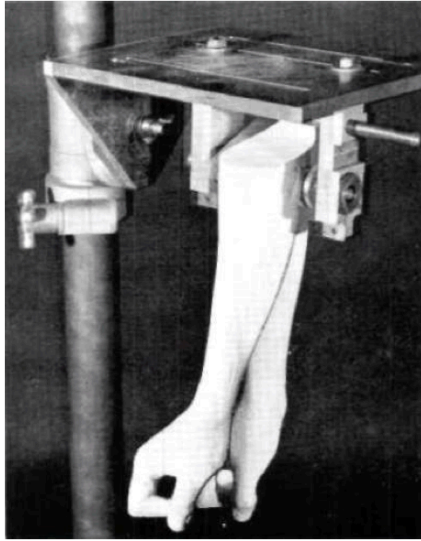
$$W = D \cdot \Delta R \div d_w$$

where  $D$  is the distance between the supports,  $d_w$  the distance the segment's centre of mass moved, and  $\Delta R$  the change in the reading.



A force plate reads the same quantity continuously, so the measurement can be made in several postures at once.

## *Cadaver dissection*



Segment a cadaver at the joints, weigh each piece, balance it to find its centre of gravity, and swing it as a pendulum to find its moment of inertia.

It is the only method that measures all three directly, and almost every number in the table traces back to it.

**Dempster's 1955 sample was eight elderly male cadavers.** Those eight people are the source of the segment masses used for every subject in most laboratories today — children, athletes, women, amputees. The table is the best we have and it is not a measurement of the person in front of you.

## Photogrammetry

Photograph the body in two planes. Measure each segment's length, width and depth from the photographs.

Treat the segment as a stack of **elliptical cylinders**, which gives a volume. Multiply by an assumed **density** and you have a mass, a centre of gravity and a moment of inertia.

The density is the catch. It is taken from the same cadaver work the method was meant to replace — the density column is the last one in Winter's table. The geometry is the subject's; the density is still Dempster's.

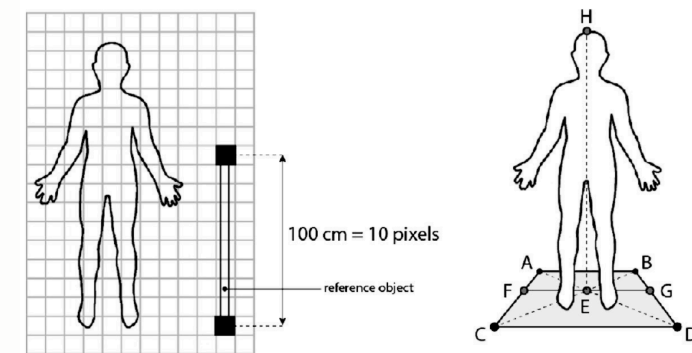


Figure 1 | Calculating centimetres per pixel using a vertical reference object (left) or a square mat (right). Using the corner points A, B, C and D the intersection point E is calculated first, followed by the distance F - G in pixels. Since F - G is also known in centimetres, the centimetres per pixel ratio is obtained. To determine total body height only point H has to be manually selected (point E is calculated automatically)

R. K. JENSEN

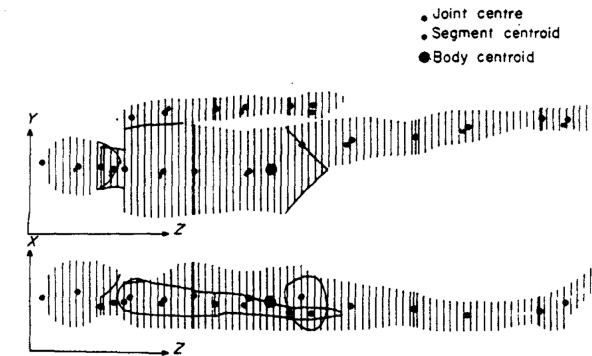


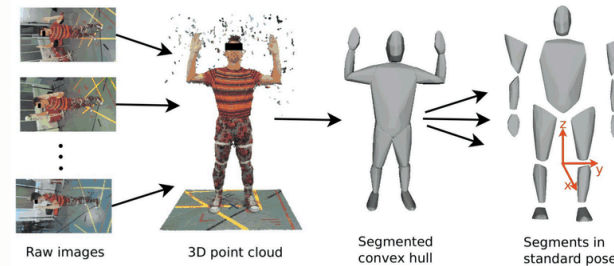
Fig. 1. The sectioned body.

## Photogrammetry, done properly

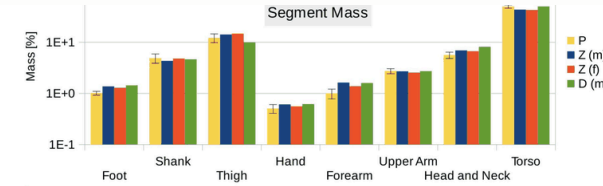
### Subject-specific body segment parameter estimation using 3D photogrammetry with multiple cameras

Kathrin E. Peyer, Mark Morris and William I. Sellers

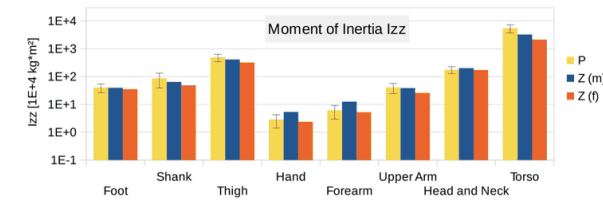
Faculty of Life Sciences, University of Manchester, Manchester, United Kingdom



**Figure 2** Image processing work flow. Images from the RPI scanner are converted to 3D point clouds which are then scaled and segmented manually. Subsequently, convex hulling is used to produce a surface mesh around each body segment.



**Figure 3** Segment mass (as % of body mass). P, Average value of all six participants (error bars show standard deviation). Foot mass adjusted by a factor of 0.51 to compensate for volume overestimation due to wearing shoes. Z(m), Male average values reported by Zatsiorsky; Z(f), Female average values reported by Zatsiorsky (Leva, 1996; Zatsiorsky, 2002); D(m), Male average values by Dempster (via Zatsiorsky) (Dempster, 1955; Zatsiorsky, 2002).



**Figure 4** Moment of inertia in ( $10^4 \text{ kg m}^2$ ) P, Average value of all six participants (error bars show standard deviation). Foot moment of inertia adjusted by a factor of 0.51 to compensate for volume overestimation due to wearing shoes. Z(m), Male average values reported by Zatsiorsky; Z(f), Female average values reported by Zatsiorsky (Leva, 1996; Zatsiorsky, 2002). The definition of the coordinate system is shown in Fig. 2.

- Many cameras rather than two, and a **3D point cloud** rather than a pair of silhouettes.
- The hull is segmented and each segment's volume taken directly — no elliptical assumption at all.
- The bar charts are the point: these masses and moments of inertia against the published values, per subject. Mostly close. Not always.

## DEXA

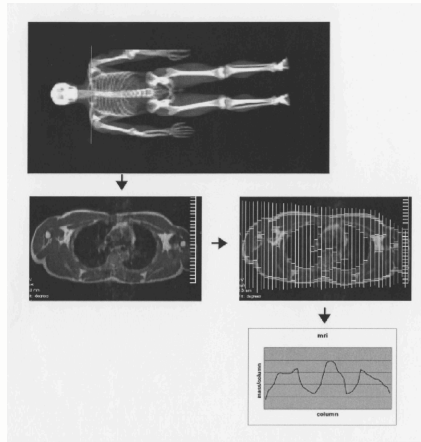
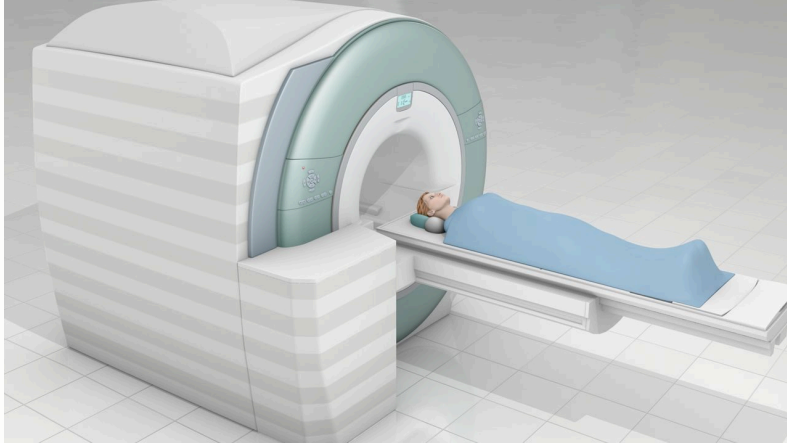
Dual-energy X-ray absorptiometry passes a **high and a low energy beam** through the body. Different tissues attenuate the two beams by different amounts, so bone, lean tissue and fat can be separated.

The output is a map of mass per unit area. Draw a boundary around a segment and its mass follows by summing — no density assumption and no geometric model.

It is a *projection*, so what comes out is mass per unit area of a two-dimensional image. Moments of inertia still need an assumption about the third dimension.



## MRI



MRI uses strong magnetic fields and radio waves rather than ionising radiation, and images **slice by slice**.

Because the slices are genuinely three-dimensional, both the centre of gravity and the **moment of inertia** of a segment can be computed directly from the images, with no geometric model and no assumed density.

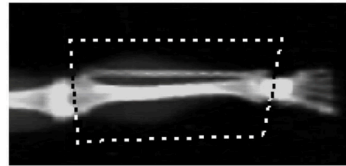
This is the only method on the list that measures everything the table contains, for the actual person. It is also the one nobody can afford to run on a hundred subjects, which is why the 1955 table is still in use.

## *An elliptical model, validated against DEXA*

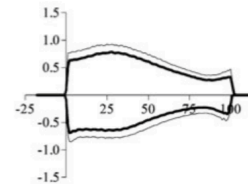
### Body segment parameter estimation of the human lower leg using an elliptical model with validation from DEXA

Jennifer L Durkin<sup>1</sup>, James J Dowling

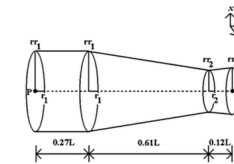
1. Scan Leg



2. Quantify mass distribution from image



3. Compare mass distribution to Geometric solid



4. Use predictive equation to determine body segment parameters from simple measures

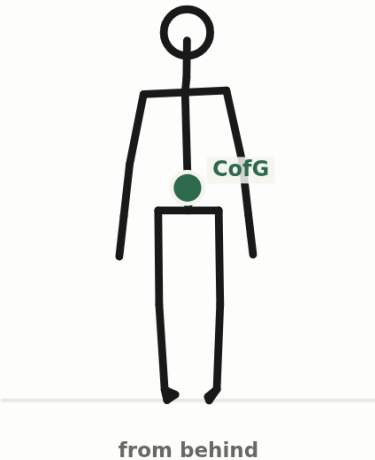
$$r_1 \sqrt{2 \left( \frac{C_1}{2\pi} \right)^2 - r r_1^2}$$

- Scan the segment, get the real mass distribution, then ask what **simple solid** reproduces it.
- This is how a method that costs a thousand dollars a subject becomes a method that costs nothing — once.
- Fit a predictive equation so that a future subject needs only a tape measure, not a scanner.

# A body that keeps changing shape

Every limb movement moves the whole

## Move a limb, move the whole



the whole body’s centre of gravity

|                          |         |
|--------------------------|---------|
| sideways                 | 0.0 cm  |
| upward                   | 0.0 cm  |
| height above the floor   | 88.6 cm |
| as a per cent of stature | 55.1 %  |

one arm is 2.75 kg, which is 5.0% of this person — a small mass moved a long way  
the shift is  $\Delta(m \cdot r) \div \Sigma m$ : the moved segment’s mass times how far it went, divided by the whole body

Measured · Subject16 · the starting pose is this person’s recorded standing trial; every bend is Winter’s table run again

Right arm

Right leg

down

neutral

Left arm

Lean the trunk

down

upright

From behind

From the side

Show the segments

Hide them

Lift one arm and the centre of gravity moves **toward it** — that is his rule, and here it is a consequence rather than a claim. Lift both and the sideways shift cancels while the upward one doubles. Lean the trunk and the shift is far larger, because the trunk is half the body.

## *Why it moves, in one sentence*

Lifting the left arm does not change how much you weigh. It changes the **moment arm** of the arm's weight about the midline.

The arm's weight now makes a larger moment to the left than before, and the only way for the moments to balance again is for the centre of gravity to move **toward** it.

$$\Delta r_{\text{cofg}} = m_{\text{limb}} \Delta r_{\text{limb}} \div \Sigma m$$

Read that as a rule of thumb: the whole body moves by the fraction of body mass that moved, times how far it went. An arm is about 5% of you, so moving it 40 cm moves your centre of gravity about 2 cm.

The general rule, from his slide:

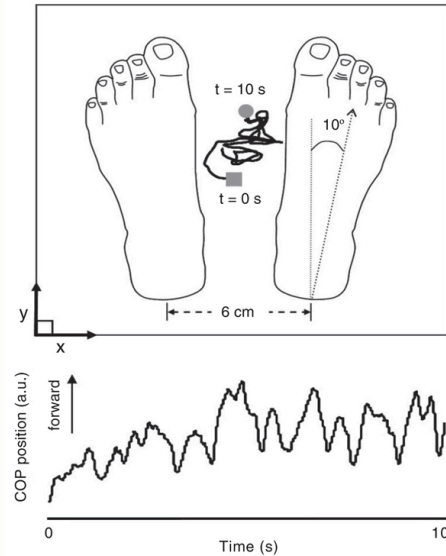
- the centre of gravity shifts **in the direction** of the limb that moved;
- how far depends on the **mass** of that limb and the **distance** it moved — that is, on the torque it caused.

And the consequence that matters for everything after this lecture: because the body changes shape constantly while walking, running or jumping, the centre of gravity is a **moving target**, not a landmark.

# Centre of gravity, centre of pressure

Two points that are easy to confuse and are never the same

## Centre of pressure



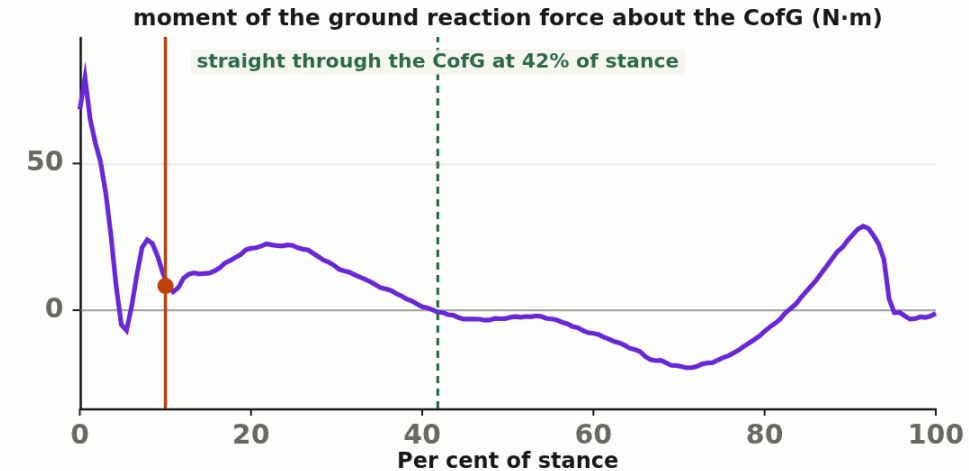
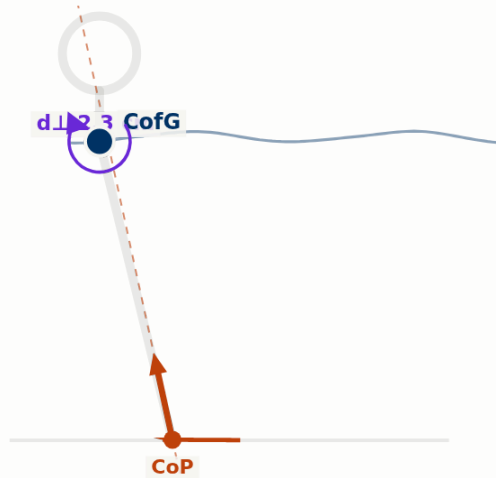
**Centre of pressure** — the point on the supporting surface where the resultant vertical force can be taken to act.

- It is a property of the *contact*, not of the body.
- On one foot, it lies under that foot. On two, it lies somewhere between them, according to how the weight is shared.
- It can never leave the base of support. The centre of gravity can, and that is what falling is.

A shift of the centre of pressure is used as an indirect measure of postural sway — but it is measuring what the *muscles are doing*, not where the body is.

# Angular Kinetics 2

*The force acts here, the body turns about there*



at 10% of stance — 8.3 N·m, turning the body backwards: the step is catching the fall

Measured · Subject16 · 55 kg · plate at 240 Hz, the authors' own centre of mass · the record opens a few milliseconds into contact

Through  
stance

10%

Show the CoP path

Hide it

► Roll through stance

The force acts at the **centre of pressure** and the body turns about its **centre of gravity**, and those are two different points — so there is a moment. Watch it change sign at midstance, where the line of action passes straight through the centre of gravity. Before that the step is catching a fall; after it, the body is falling into the next one.

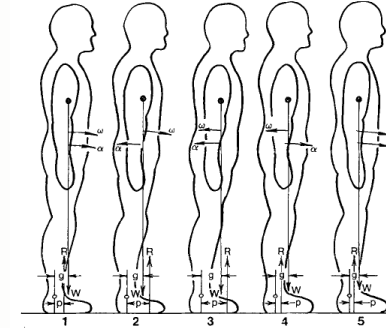
# Angular Kinetics 2

## Quiet standing

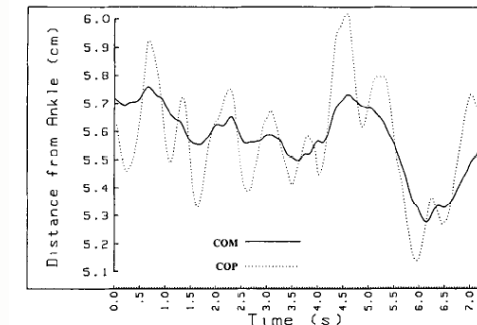
Stand still and the centre of gravity drifts. The centre of pressure moves **around** it, crossing from one side to the other, creating alternating “righting” moments.

Winter’s record is the classic demonstration: the two traces on one axis, over seven seconds.

The centre-of-pressure trace has a **higher frequency** and a **larger amplitude** than the centre-of-gravity trace. That is not an artefact and it is not noise. It is the control signal, and the next slide shows why it has to be bigger.



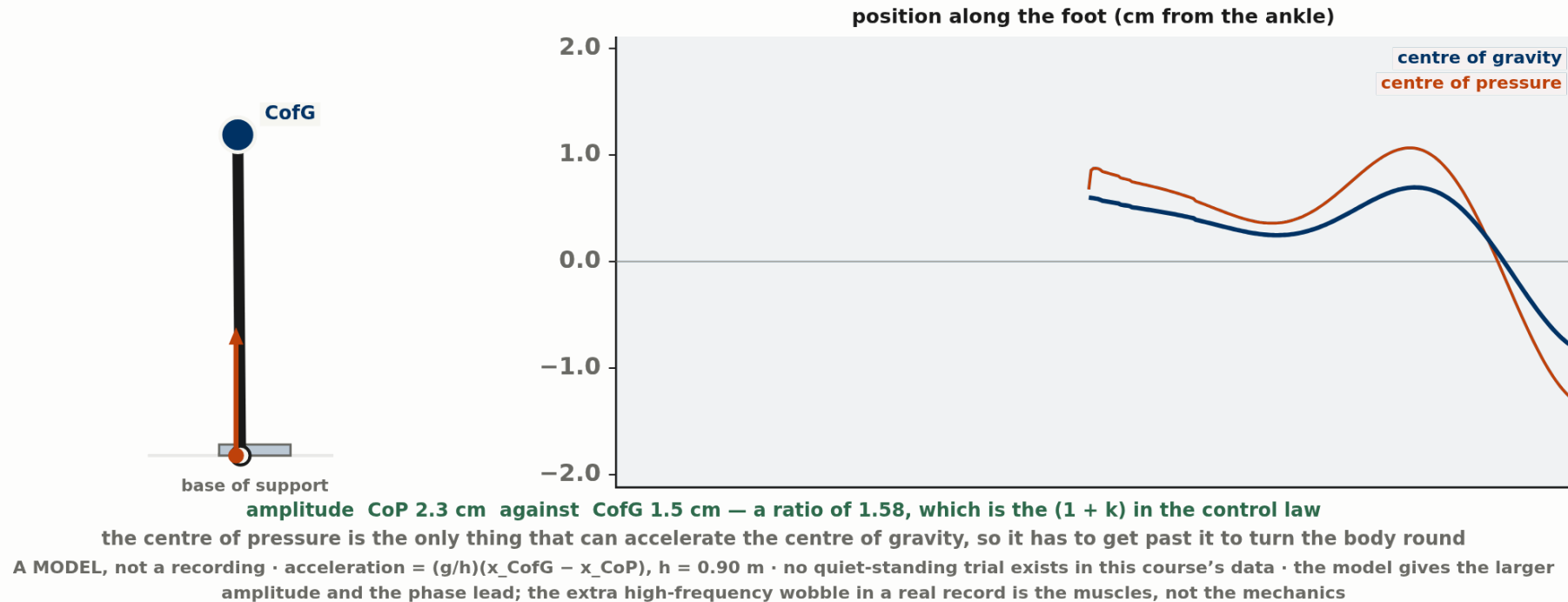
**Figure 1.** A subject swaying back and forth while standing quietly on a force platform. Five different points in time are described, showing the centre of gravity (g) and the centre of pressure (p) locations along with the associated angular accelerations ( $\alpha$ ) and angular velocities ( $\omega$ ). See text for detailed description.



**Figure 2.** A 7-s record showing simultaneous centre-of-gravity and centre-of-pressure fluctuations for a subject in quiet stance. Centre-of-pressure excursions oscillate either side of the centre of gravity and have a higher frequency and greater amplitude.

# Angular Kinetics 2

## Why the centre of pressure has to overshoot



How far past  
the CofG you  
put the force

× 1.50

How far you  
lead the sway

0.56

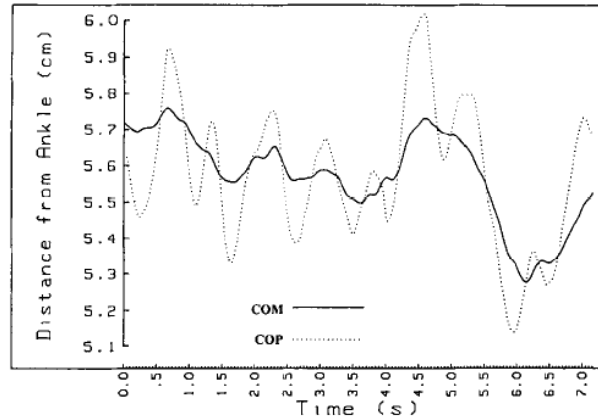
Let it balance itself

You hold it up

Reset

Standing is an inverted pendulum, and the centre of pressure is the only handle on it. Watch the two traces: the **centre of pressure swings further, and gets there first**, crossing from one side of the centre of gravity to the other — which is what Winter's record shows. Then press **You hold it up** and move the pointer across the figure to put the force where you want it. Most people fall over in about two seconds.

## The base of support



**Figure 2.** A 7-s record showing simultaneous centre-of-gravity and centre-of-pressure fluctuations for a subject in quiet stance. Centre-of-pressure excursions oscillate either side of the centre of gravity and have a higher frequency and greater amplitude.

The **base of support** is the area enclosed by the parts of the body in contact with the ground, including the space between them.

Balance is one condition: keep the centre of gravity's vertical projection **inside** that area. The centre of pressure is the tool for doing it, and it can only ever act within the base.

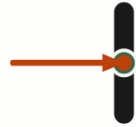
- Feet apart — a wider base, more room before a correction is needed.
- A cane — a *much* larger base, which is most of what a cane is for.
- Up on the toes — a base a few centimetres long, which is why it is hard.

# Forces through it, and forces that miss

Why anything spins at all

# Angular Kinetics 2

*Through the centre, and off it*



what the flick does

$$\mathbf{F} = m \mathbf{a}$$

the centre of gravity accelerates to 7.5 m/s, and nothing else happens at all

$$\mathbf{M} = \mathbf{F} d_{\perp}$$

the force passes through the centre of gravity, so  $d_{\perp} = 0$ , there is no moment, and it will not turn however hard you push

The green line is straight, and at the same speed, wherever you flick it. An off-centre push ADDS a spin; it does not take anything away from the translation.

set where and how hard, then press Flick it

A uniform rod,  $L = 22 \text{ cm}$ ,  $m = 12 \text{ g}$  · one impulse applied square to it, then free flight at 3% speed

Where you  
flick it



centre

How hard



0.09 N·s

▶ Flick it

Flick it **through the centre** and it slides without turning. Move the flick off-centre, leaving everything else alone, and it slides *exactly as before* and spins as well — and the spin is about the centre of gravity, not about the point you pushed. The green path is the proof: it never bends, and it never slows down.

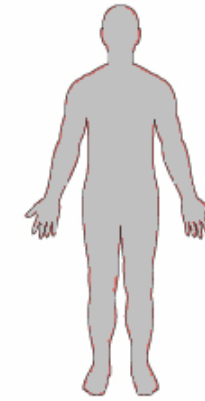
## *The two cases, stated*

A force **through** the centre of gravity:

- changes the **linear** motion of the object;
- produces no moment about the centre of gravity, because  $d_{\perp} = 0$ ;
- so there is no change in angular motion at all.

A force **not through** the centre of gravity:

- changes the linear motion in *exactly the same way*;
- *and* produces a moment  $M = F d_{\perp}$ ;
- so the object translates and rotates — and it rotates about its centre of gravity.



Newton's second law in its two forms, side by side:  **$F = ma$**  never mentions where the force was applied, and  **$M = I\alpha$**  is the only equation that does. That is the entire reason a free object spins about its centre of gravity and nothing else.

An unconstrained body has no axle. The only point it *can* rotate about is the one for which the weights balance — which is the point this whole lecture has been about.

# Angular Kinetics 2

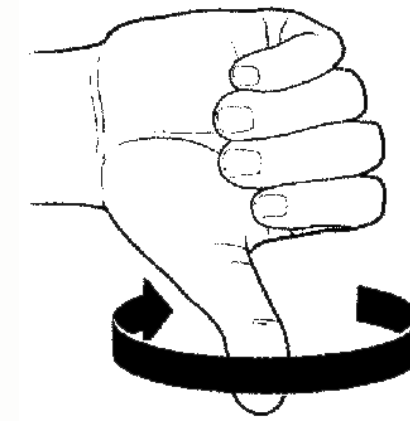
## Example 1 — a force at right angles

A 75 N force acts perpendicular to a post, 3 m from the axis at **O**.  
Find the moment about O.

**Magnitude.** The force is already perpendicular, so the moment arm is the distance given:

$$M_O = F d_{\perp} = (75 \text{ N})(3 \text{ m}) = \mathbf{225 \text{ N}\cdot\text{m}}$$

**Direction.** By the right-hand rule the moment vector points orthogonally to the page, with a sense going *into* it — which in the usual sign convention makes it **negative**.



This is the same example as Angular Kinetics 1, slides 8 to 10, and the deck for that lecture has it as an interactive figure. If it is fresh, move through these four slides quickly.

## *Example 2 — the same force, at 30°*

### **Solution A — resolve the distance.**

Find the perpendicular distance from the line of action to the axis:

$$d = (3 \text{ m})(\cos 30^\circ) = 2.6 \text{ m}$$

$$M_O = F d_{\perp} = (75 \text{ N})(2.6 \text{ m}) = \mathbf{195 \text{ N}\cdot\text{m}}$$

### **Solution B — resolve the force.**

Only the component perpendicular to the post contributes:

$$(75 \text{ N})(\cos 30^\circ) = 65 \text{ N}$$

$$M_O = (65 \text{ N})(3 \text{ m}) = \mathbf{195 \text{ N}\cdot\text{m}}$$

Two routes, one answer. Resolve the *distance* or resolve the *force* — never both, which is the commonest error on this question.

## *What to take away*

- **One sum does all of it.**  $\Sigma(m \cdot r) \div \Sigma m$  locates the centre of gravity in one dimension, and you run it once per axis for the others. There is no separate two-dimensional formula.
- **It is the balance point because the moments cancel there** — that is not a second fact about it, it is the same fact rearranged.
- **Segment masses are not measured on your subject.** They come from eight elderly male cadavers by way of Dempster, and every centre-of-gravity and inverse-dynamics result in this course inherits that.
- **The centre of gravity moves whenever the body changes shape**, by the fraction of body mass that moved times how far it went. The trunk usually wins.
- **Centre of pressure is not centre of gravity.** One is a property of the contact and can never leave the base of support; the other is a property of the body and leaving the base is exactly what falling is.
- **The centre of pressure has to overshoot the centre of gravity**, because it is the only thing that can accelerate it. That is why its trace is the bigger, busier one.
- **A force through the centre of gravity translates; a force anywhere else translates the same and rotates as well.**  $F = ma$  does not care where you pushed. Only  $M = Fd_{\perp}$  does.

# Angular Kinetics 2

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Next up: angular kinetics 3 — moment of inertia, angular momentum and how a diver changes her rate of spin without changing her angular momentum · Measured data: Lencioni et al., two subjects, standing and level walking · Segment parameters: Winter, Table 4.1, after Dempster